

A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 04

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two reasons*), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	<i>1 mark for each point.</i> • Landfill fell from 79% to 14% (large decrease) (1) • Incineration rose from 9% to 42% and recycling / composting rose from 11% to 43% (now the largest) (1)	2
01.2	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Organic matter decays anaerobically releasing methane (and CO₂) - greenhouse gases (1) • Toxic leachate may leak into groundwater / rivers from poorly managed sites (1) • Uses large areas of land; loss of habitats / farmland; contamination prevents later use (1) • Resource value of metals, glass, plastics is lost (1) • Transport noise / congestion / litter / pests / odour (1)	3
01.3	<i>Any 4 from the points below - 1 mark each, maximum 4.</i> • Polymer / clay liner to prevent escape of leachate (1) • Collection and treatment of leachate (1) • Collection of methane and use as a fuel (1) • Impermeable cap (clay and polymer) when complete to stop water entering (1) • Regular covering with soil (pests, odour); deodorising sprays (1) • Perimeter fence to trap wind-blown litter (1) • Separation and recording of waste types; reactive / flammable wastes not dumped together (1)	4
01.4	<i>1 mark.</i> • Mixed / contaminated / dirty wastes (plastics, dirty paper) are difficult and costly to separate and recycle but can be burnt with energy recovery (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 280-282; AQA spec 3.4.3.2.12.	

Question	Marking guidance	Marks
02.1	<i>Correct answer only.</i> • Its total annual discharge (1)	1
02.2	<i>1 mark for each point.</i> • Public supplies cannot match consumer demand / the water is often cleaner than rivers (1) • It also reduces urban flooding after heavy rain; surplus can be used for aquifer recharge (1)	2
02.3	<i>1 mark for each point.</i> • The reservoir is created in the estuary so no farmland / settlements need to be flooded (1) • But it changes important inter-tidal habitats, obstructs shipping and can be polluted by activities anywhere in the river's catchment (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 122-123; AQA spec 3.2.2.5.	
03.1	<i>1 mark for the correct answer.</i> • $4.2 / 0.7 = 6$ (1)	1
03.2	<i>1 mark for each point.</i> • Long summer days (almost 24 hours of daylight) so the total energy received is high, even though the Sun is low (1) • In winter days are very short / the Sun does not rise, so light levels are extremely low (1)	2
03.3	<i>1 mark for each point.</i> • Reflectors can only focus parallel rays of direct sunlight onto the absorber (1) • Clouds, smoke and dust scatter the light so it cannot be focused and intensified (1)	2
03.4	<i>1 mark.</i> • Very large areas of collectors are needed to harness significant amounts of energy (and the changing angle of incidence reduces it further) (1)	1

Question	Marking guidance	Marks
03.5	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Problems: intermittency (daily and seasonal cycles), unpredictable cloud cover, low energy density, changing angle of incidence, land requirements, water for cleaning in deserts, toxic wastes in manufacture (silicon tetrachloride, cadmium), efficiency loss at high temperature • Photothermal: evacuated tubes reduce conduction / convection losses; thermal stores (water, sand, concrete, molten salt) allow later use; passive solar architecture, brise soleil, light tubes; heat pumps deliver up to four times the energy used • Photovoltaic developments: multijunction cells absorbing more wavelengths, anti-reflective / moth-eye textured surfaces (smooth cells reflect 30%), PV-thermal hybrids cooling the cell, transparent PV in windows, heliostats keeping the optimum angle, self-cleaning nanohydrophobic surfaces • Concentrating solar power with molten-salt storage to 550 °C so steam turbines run when needed - overcomes intermittency • Integration: grid-connected solar farms, rooftop space avoiding land conflicts, batteries, pumped storage, links to energy storage and demand management • Evaluation: strong in sunny deserts / low latitudes, weaker in high latitudes in winter; cost falling; manufacturing impacts; no single technology solves all problems; justified conclusion	9
	Model answer - Level 3 (9 marks)	
	Solar power's problems are intrinsic: it is intermittent with daily and seasonal cycles, cloud makes it unpredictable and scatters light so it cannot be focused, its low energy density demands large collector areas, the angle of incidence keeps changing, cleaning in deserts uses scarce water, manufacturing PV produces toxic wastes such as silicon tetrachloride and cadmium, and PV efficiency falls as cells heat up.	

Question	Marking guidance	Marks
	New technology addresses most of these. Evacuated-tube photothermal panels cut conduction and convection losses, and thermal stores of water, sand, concrete or molten salt hold heat for later; passive solar design, brise soleil and light tubes reduce demand; heat pumps deliver up to four times the energy used to run them. Multijunction PV cells absorb more wavelengths, moth-eye anti-reflective surfaces stop the 30% of light a smooth cell reflects, PV-thermal hybrids cool the cell and use the heat, transparent PV turns windows into generators, heliostats keep panels at the optimum angle and nanohydrophobic surfaces self-clean. Concentrating solar power with molten salt stored at 550 °C runs steam turbines when electricity is wanted, directly overcoming intermittency, and grid-connected farms, rooftops that avoid land conflicts, batteries and pumped storage integrate the output.	
	Even so, no technology changes the fact that winter insolation at high latitudes is tiny, that CSP needs cloudless skies, or that manufacturing has impacts. Solar can dominate supply in sunny, low-latitude regions and contribute strongly elsewhere in summer, but overcoming its problems fully depends on storage and a diverse energy mix rather than on solar technology alone.	
	Weaker answer - Level 1 (2 marks)	
	Solar power has problems because it does not work at night or when it is cloudy, and solar panels are not very efficient. New technologies can help. Better solar panels are more efficient. Solar power can be stored in batteries for use at night. Mirrors can concentrate the sunlight to make it hotter. So new technologies can overcome most of the problems with solar power.	
	What makes the difference	
	• Level 1 identifies two problems and three vague technologies; Level 3 names each technology and explains which problem it solves and how.	
	• Level 3 acknowledges what technology cannot fix (high-latitude winters, cloud for CSP, manufacturing impacts) - the 'extent' in the question.	
	• Numbers (30% reflection, 550 °C, four-fold heat-pump output) show detailed knowledge applied to the argument.	
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 199-203; AQA spec 3.3.4.1.	
04.1	<i>Correct answer only.</i> • Organic (methyl) mercury (1)	1
04.2	<i>1 mark for each point.</i> • Anaerobic microbes converted the inorganic mercury into organic methyl mercury (1) • Methyl mercury is liposoluble so it is absorbed, crosses into the brain and biomagnifies along the food chain to fish eaten by people (1)	2
04.3	<i>1 mark for each point.</i> • Most heavy metals are less soluble at high pH (1) • So they remain immobile and are not leached into water or absorbed by organisms (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 275-279; AQA spec 3.4.3.2.11.	

Question	Marking guidance	Marks
05.1	1 mark for each point. - The amount of energy contained in a given mass of the resource (or harnessed per kg of equipment for renewables) (1) - High energy density means smaller quantities are needed so storage and transport are easier and cheaper (eg 75 t of aviation fuel carries a 400 t aircraft 5600 km) (1) - High energy density fuels reach high temperatures needed for smelting / steam turbines (1)	3
05.2	A correct final answer (with unit where asked) scores full marks. If the answer is wrong, award 1 mark for each correct step of working shown. $500\,000 / 28 = 17\,857\text{ kg}$ (1) $= 17.9\text{ tonnes}$ (accept 18; book quotes 13 t for 0.7% enriched fuel) (1)	2
05.3	1 mark for each point. - The electricity they generate can power high-energy-requirement technologies (1) - Electricity can be converted to high energy density secondary fuels, eg hydrogen by electrolysis, or stored (batteries, pumped storage) (1)	2
05.4	1 mark for each point. - Intermittent: not available all the time, eg tidal (but the times are predictable so alternatives can be planned) (1) - Unpredictable: intermittent and the timing / amount cannot be forecast accurately, eg wind, solar (1)	2
05.5	1 mark. Gaseous natural gas has a very low energy density by volume; LNG at $-160\text{ }^{\circ}\text{C}$ has 160 times the energy density so a ship can carry a worthwhile amount (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 175-178; AQA spec 3.3.2.	
06.1	Correct answer only. A waste is assumed to cause pollution until research shows it is unlikely to do so (1)	1
06.2	1 mark for each point. - If the cost of preventing damage is less than the (penalty for) damage there is a financial incentive to prevent pollution (1) - If polluters can escape their responsibilities / penalties are low, there is little incentive to comply (1)	2
06.3	Any 2 from the points below - 1 mark each, maximum 2. - Marine discharges where currents dilute and disperse emissions (1) - Emissions downwind of urban areas (1) - Not discharging waste onto permeable rock above an aquifer (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 243-244; AQA spec 3.4.3.1.	
07.1	1 mark for each point. - UV light splits diatomic oxygen (O_2) into two monatomic oxygen atoms (O) (1) - Monatomic oxygen reacts with diatomic oxygen to form ozone (O_3) (1) - UV light also splits ozone into O_2 and O; the reactions form a dynamic equilibrium which absorbs UV (converting it to chemical energy / heat) (1)	3

Question	Marking guidance	Marks
07.2	<i>Up to 6 marks in total; award marks in proportion to the completeness of each point.</i> • Useful: boiling points near ambient temperature so easily liquefied under low pressure (aerosol propellants, fridge / air-conditioning coolants); dissolve grease but do not damage electrical components (solvents); non-flammable; non-toxic (up to 2) • Persistence: chemically stable / unreactive in the troposphere so they remain long enough to be carried up to the stratosphere (up to 2) • Dissociation by UV: in the stratosphere higher UV levels break the carbon-chlorine bonds releasing chlorine free radicals (up to 2) • Chlorine reacts with monatomic oxygen (and ozone), preventing ozone formation; chlorine is regenerated so one atom destroys many ozone molecules; bromine / iodine act similarly (up to 2)	6
07.3	<i>1 mark for each point.</i> • UVB energy is absorbed by cells and breaks up biological molecules / DNA causing mutations, skin cancer, cataracts, sunburn (human health) (1) • Damage to leaf tissue and reduced photosynthesis / crop yields (plants) (1) • Damage to marine algae, phytoplankton, corals and planktonic larvae (marine organisms), reducing productivity and food webs (1)	3
07.4	<i>1 mark.</i> • An area where the total ozone column falls below 220 Dobson Units (normal is about 300 DU) - a level never recorded before CFC pollution (1)	1
07.5	<i>1 mark.</i> • Concentration varies with altitude and the stratosphere is not normally sampled; Dobson Units measure the total column thickness instead (1)	1
07.6	<i>1 mark.</i> • The British Antarctic Survey at Halley detected higher UV at ground level, showing ozone above had decreased (satellites above the stratosphere cannot measure downward UV directly) (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 108-109; AQA spec 3.2.1.3.	
08.1	<i>Correct answer only.</i> • Oxygen (1)	1
08.2	<i>1 mark for each point.</i> • A change to one process causes other processes to alter as a direct result (chains of linked processes / feedback) (1) • There is still a lot that is not understood about how atmospheric processes work (1)	2
08.3	<i>1 mark for each point.</i> • The surface warmed in the tropics heats the air above; warm winds (south-westerlies from the Caribbean) carry heat energy to higher latitudes (1) • Winds carry water vapour (evaporated from warm oceans) to areas that would otherwise receive little precipitation (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 86-88; AQA spec 3.2.1.1.	

Question	Marking guidance	Marks
09.1	<i>1 mark for each point.</i> • Soil loss increases as vegetation cover decreases: forest lowest (0.5), bare soil highest ($60 \text{ t ha}^{-1} \text{ yr}^{-1}$) - over 100 times more (1) • Ploughing downslope loses nearly three times as much as contour ploughing for the same crop (1)	2
09.2	<i>Any 4 from the points below - 1 mark each, maximum 4.</i> • Leaves / leaf litter intercept raindrops so soil particles are not dislodged by rain splash (1) • Roots bind soil particles together (1) • Vegetation acts as a windbreak reducing wind velocity and the kinetic energy available to carry particles (1) • Plants increase infiltration (root channels, organic matter) so less surface runoff (1) • Organic matter / humus from vegetation binds particles into stable peds (1)	4
09.3	<i>1 mark for each point.</i> • Furrows running downslope act as channels / gulleys that increase the velocity (kinetic energy) of runoff (1) • Furrows along the contour trap the water so it loses energy, deposits particles and infiltrates (1)	2
09.4	<i>1 mark for each point.</i> • In the dry season the soil (especially low-clay soils) is loose and dry with little cohesion so wind blows it away (1) • In the wet season heavy rain causes rain splash and surface runoff when infiltration capacity is exceeded (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 162-164; AQA spec 3.2.5.2.	
10.1	<i>1 mark for each point.</i> • t-test (or Mann-Whitney U if not normally distributed) (1) • Comparing two means of counted / measured data from two independent samples (1)	2
10.2	<i>1 mark for each point.</i> • The difference is statistically significant (less than 1% probability that it arose by chance) (1) • The null hypothesis (no difference between fields) is rejected; the organic field has significantly more earthworms (1)	2
10.3	<i>1 mark for each point.</i> • A significant difference does not show a causal relationship (1) • Other variables (soil type, moisture, pH, cultivation history) may differ between the two fields and cause the difference (1)	2

Question	Marking guidance	Marks
10.4	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Scientific methodology: identifying the topic, reviewing existing knowledge, formulating a testable hypothesis and null hypothesis, designing the study, analysing and drawing conclusions, planning further research • Sample location: random (grid coordinates / numbered sites) or systematic (transects, fixed intervals) to avoid bias; interval chosen so variation is detected without excess data • Sample timing: diurnal, seasonal, weather-related and long-term changes; sampling on different occasions; interval matched to rate of change • Sample size and number: larger and more samples for non-homogeneous variables; running mean and standard deviation show when the mean is reliable; preliminary study to decide • Standardisation and control of variables; accuracy, precision, reliability, representativeness; anomalies and repeats; calibration of equipment; appropriate techniques and risk management • Statistical analysis: choosing the test (Spearman, chi-squared, t-test, Mann-Whitney), significance and p-values, error bars, correlation vs causation • Evaluation: trade-offs between cost, time, disturbance and reliability; limitations of sub-sampling; justified conclusion	9
	Model answer - Level 3 (8 marks)	
	A valid investigation begins with a clear, testable hypothesis and a null hypothesis, informed by existing knowledge, so that the data collected can actually accept or reject it. The first design decision is sample location: random sampling (grid coordinates or numbered sites) or systematic sampling along transects avoids the bias that would come from choosing convenient sites, and a preliminary study shows the interval needed to detect variation without collecting unnecessary data.	

Question	Marking guidance	Marks
	Timing matters because many variables change diurnally, seasonally or with weather, so samples on different occasions are needed and the interval must match the rate of change. Sample size and number depend on how variable the population is: a running mean shows when enough samples have been taken and the standard deviation shows the scatter around the mean. Techniques must be standardised so results from different sites, times or researchers are comparable, other variables controlled or monitored, and equipment calibrated so that precise readings are also accurate; anomalies should be repeated rather than ignored.	
	Finally the analysis must be chosen to suit the data - Spearman's rank for correlation, chi-squared for frequencies, a t-test for two means or Mann-Whitney U for medians - so that significance ($p < 0.05$) can be assessed, remembering that a significant correlation does not prove causation. Every decision trades reliability against cost, time and disturbance; sub-samples can only estimate the whole, so the most valid study is the one whose sampling design makes its limitations explicit.	
	Weaker answer - Level 2 (5 marks)	
	When planning fieldwork you need a hypothesis. You should choose sites randomly so it is not biased, for example using random numbers. You need to take enough samples and repeat them so the results are reliable. The method should be the same each time. You should measure accurately with calibrated equipment. Then you do a statistical test such as a t-test to see if the results are significant. If p is less than 0.05 the result is significant. This makes the conclusions valid.	
	What makes the difference	
	• Level 2 is sound but generic: it says 'enough samples' where Level 3 explains how a running mean and standard deviation tell you how many.	
	• Level 3 covers timing and the choice between statistical tests; Level 2 mentions one test only.	
	• The evaluation is missing from Level 2 - there is no discussion of trade-offs or of correlation vs causation, so it cannot reach the top level however accurate it is.	
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 390-393, 420-423; AQA spec 3.7.1, 3.7.2.3.	

Question	Marking guidance	Marks
11.1	Discuss the extent to which energy conservation technologies could reduce the need for new energy supplies in a developed country. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Scale of the problem: how energy is used (transport, buildings, industry, water); links between affluence and per capita use • Transport: bulk / integrated transport, ATM, vehicle design (aerodynamics, mass, tyres, KERS, combustion efficiency), low embodied energy, design for end of life, driver behaviour, electric vehicles • Buildings: orientation and passive solar, SA:V, thermal mass, low embodied energy materials, earth sheltering, glazing (low-e, inert gas, vacuum), insulation optimum, heat exchangers, automatic ventilation, occupancy sensors, thermostats, low-energy appliances, behaviour • Industry: heat recovery and exchangers, insulation and high-volume storage, CHP, integrated manufacture, recycling and mass reduction, high-voltage grid, peak shaving, ICT management • Limits: rebound effect, cost and payback, existing building stock, embodied energy of replacements, behaviour • Evaluation with quantitative examples; comparison with the cost of new supply; conclusion that conservation is the cheapest first step but cannot alone meet demand growth Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 170-174, 225-235; AQA spec 3.3.4.2, 3.3.1.	25

Question	Marking guidance	Marks
11.2	Discuss how knowledge of the properties of energy resources can be used to reduce the severity of their impact on the environment. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Properties: energy density, renewability, abundance, location, intermittency / predictability, storage, transport, conversions, technology level, environmental impacts • Fossil fuels: choosing lower-impact fuels (gas over coal), desulfurisation, CCS, efficient combustion, siting power stations near cooling water and coal ports; new extraction technologies (directional drilling, secondary / tertiary recovery) and their impacts (fracking, tar sands) • Nuclear: high energy density reduces transport; waste properties (half-life) determine storage; thorium and fusion prospects • Renewables: matching resource to location (wind, HEP, solar, wave, tidal, geothermal, biofuel siting); intermittency managed by storage (pumped HEP, batteries, hydrogen, P2G, V2G), grid management and diversity; low-impact designs (helical turbines, VAWT, tidal lagoons) • Energy conservation reduces demand: transport, buildings, industry (CHP, heat exchangers) • Impacts before and during use; life-cycle / embodied energy • Evaluation: no resource is impact-free; knowledge allows the best mix and mitigation; judgement Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 175-235; AQA spec 3.3.2-3.3.4.	25